Models in Keras

Keras comes with several built-in models with pre-trained weights, which can be used directly. The only thing required to be done is to re-size your images beforehand due to the fully connected layers at the end that forces the input size to be fixed. It provides a Model class that you can use to create a model from your created layers. It requires you to just specify the input and output layers. There are two types of built-in models that are available in Keras: sequential models and models that are created with the functional API. In addition, you can also create custom models that can be used to define your own forward-pass logic. All models have the following common properties:

- Model layers that are a flattened list of the layers, which comprise the model graph.
- Model inputs that contain a list of input tensors.
- Model outputs that contain a list of output tensors.

Building a model in Keras

To build a Keras model, you need to carry out a few steps. First, we instantiate a model, then we add layers to it one by one. After that, we need to compile a model with a compulsory loss function and a mandatory optimiser (like Adam Optimiser). We can also use optional evaluation metrics such as accuracy. Hereafter, we fit our model to training data from the data set. The usual approach after that is to evaluate our model but, sometimes you may want to serialise the model or deploy it somewhere or start a new experiment altogether which depends on your use. This list of steps is required each time when you define a Keras model.

Sequential Models

The Sequential model is a linear stack of layers, which can be described very simply. They are created using the *keras_model_sequential ()* function. A Sequential model can be created by passing a list of layer instances to the constructor.

The Sequential model API is great for developing deep learning models in most situations, particularly when the task is not very complex. It provides a high level of abstraction and is user friendly. However, it has some limitations. It does not allow you to create models that share layers or have multiple outputs or inputs. It is difficult to define those models that have multiple unique input sources, produce multiple output destinations, or in models